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Does Digitalisation Promote the Servicification of Manufacturing? Firm-Level Evidence From China

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ABSTRACT

In this paper, we explore the digital transformation of Chinese firms and their impact on service integration within manufacturing activities in China. Using highly disaggregated firm-product level data, we examine various dimensions of the servicification in manufacturing and assess how digital adoption affects these processes. Our findings indicate that the adoption of digital technologies significantly drives Chinese manufacturing firms to generate more service outputs, either by complementing traditional manufacturing activities or by increasingly substituting manufacturing products with services. Moreover, we observe that digital adoption substantially improves the productivity and profitability of these firms, further incentivising them to expand their service offerings. Using text mining and sentiment analysis, we find that only manufacturing firms with positive attitudes toward digitalisation strategy tend to increase their service outputs. These results highlight the transformative effects of digital technology on manufacturing production, shedding new light on the evolving trends of service-led global value chain upgrading in China.

JEL Classification: F14, F23

1 | Introduction

The evolution of production modes within manufacturing firms has increasingly been affected by the integration of digital technology, shifting from the traditional manufacturing-led mode to service linkages and sectors, especially in developing countries (Kilic and Marin 2020). This change is facilitated through the integration of advanced technologies such as artificial intelligence (AI), the Internet of Things (IoT), robots, and automation. These technologies not only streamline the manufacturing operations but also enhance the service production of these firms, such as maintenance, data analytics, finance, or after-sales support (Jung and Gómez-Bengoechea 2022). This trend, known as the servicification of manufacturing, has gained traction in both developed and developing countries (Lodefalk 2013; Thangavelu

et al. 2018). Digital technology, as a critical new factor of production, has fundamentally changed the comparative advantages of developing countries in manufacturing sectors, transforming from labour-intensive manufacturing tasks to more knowledge-based, service-oriented activities (Baldwin and Forslid 2020).

Despite the rapid development of digital technology and its vital role in driving manufacturing firms to engage in the services sector, however, few studies quantify the impacts of digitalisation on the servicification of manufacturing, especially at the firm level (Thangavelu et al. 2018). This paper fills this gap by examining the impacts of digitalisation on the servicification of manufacturing of Chinese manufacturing firms. Using detailed firm-product level data, this paper captures the different dimensions of servicification of manufacturing firms and investigates

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the causal relationship between digitalisation and the integration of services within manufacturing firms.

One challenge to achieving this is to accurately measure the levels of servicification and digitalisation at the firm level. Following Lodefalk (2013), we decompose the income sources of listed Chinese manufacturing firms into product levels, enabling the identification of income derived specifically from service outputs. We define the level of servicification by the share of service-related income to the total income of the manufacturing firms, which enables us to trace the evolving dynamics of manufacturing servicification at the firm level. Furthermore, following Cusumano et al. (2015), we classify the services produced by manufacturing firms into three categories: smoothing services that facilitate the manufacturing production process, adapting services that enhance the manufacturing products' functionality, and substituting services that replace manufacturing outputs. Our analysis provides new insights into how digitalisation affects these specific categories of servicification, revealing the heterogeneous linkages between digital technology advancement and service integration in manufacturing processes.

Meanwhile, we have developed a comprehensive methodology to measure digitalisation at the firm level. Various indicators have been adopted in previous studies to capture firm-level digital technology adoption, such as the ICT adoption rate (Hall et al. 2013; Gopalan et al. 2022), the digital adoption of unusual breadth (Nucci et al. 2023), and the share of intangible assets in total assets (Tambe et al. 2020; Dhyne et al. 2020). However, these indicators specifically focus on the adoption of specific digital technologies, lacking a holistic view of the digital transformation of firms. Another branch of literature uses text mining and analysis to collect digitalisation-related texts on websites or annual reports as a proxy of digitalisation (Chen et al. 2022; Axenbeck and Breithaupt 2022; Wu et al. 2021). The advantage of this approach is to generate real-time and large-scale estimates of firms' digitalisation strategies, giving a comprehensive estimation of firms' digital technology adoption level. However, given the interactive nature of these text sources, estimation bias might arise if the disclosure is affected by firm stakeholders' preferences (Lee 2016; Li et al. 2021).

In this paper, we first use the text mining approach to analyse annual reports of Chinese-listed firms between 2000 and 2020 to identify and quantify textual references to digitalisation. The digital transformation of firms is based on the frequency of terms related to five key dimensions of digital technology, including AI, big data, cloud computing technology, blockchain technology, and digital technology application. Additionally, we use sentiment analysis to assess the emotional tone within texts of digitalisation, which enables us to further distinguish firms' attitudes toward digital adoption¹ (Shaik et al. 2023). This approach allows us to capture firms' attitudes toward digitalisation, which provides detailed information on how firms adjust their digital strategies. Lastly, considering that the references to digital technology in the text do not necessarily confirm its implementation, this paper also uses the size of intangible assets related to digital technology to capture firms' investment in digital technology as an alternative indicator of digitalisation.

Another challenge in this paper is to address the potential endogeneity between digitalisation and the servicification of manufacturing at the firm level. The integration of technology within services often blurs the line between digital inputs and service outputs. For instance, manufacturing firms may offer AI-based services that are directly supported by AI technology, making it difficult to separate the service component from the technological input. In response to this, we first identify digitalisation in this paper as the adoption of digital technologies that generate, process, and transfer information (Katz and Koutroumpis 2013). We further decompose service outputs at the product level, clarifying the distinction between digital inputs and their service applications. These approaches clarify the ambiguity between digitalisation and servicification and eliminate the endogeneity due to measurement errors.

Moreover, firms that generate more service outputs may also be more inclined to adopt advanced digital technologies, which may also lead to endogenous problems. To address this issue, this paper identifies the causal effect of digitalisation on firms' servicification trends using the city- and time-variations generated by the Broadband China pilot city policies. Broadband China was promulgated in 2013 by the central government of China and aims to develop the broadband infrastructure of pilot cities. There were 117 cities included in the pilot cities of the Broadband China Program in 2014, 2015, and 2016, respectively, which substantially increased the accessibility of firms in these regions to digital technology. Previous studies have shown that the Broadband China policy has significantly increased the digitalisation level of local firms (Zhang and Liu 2023).

Our empirical analysis first uses regional variations generated by the Broadband China reform, which was piloted in 117 of 293 cities across time, as an instrument of digitalisation. Additionally, we provide extra instruments to further reduce the possible endogeneity. The first instrument is the average digitalisation level of other firms within the same sector and city. This instrument captures the sector-level and firm-level variance in digitalisation while eliminating reverse causality by excluding the focal firm. In addition, we use the number of post offices and telephones in 1984 as instruments for digitalisation. These historical indicators are directly related to the early stages of communication infrastructure development, which are crucial elements of digitalisation, while barely affected by servicification of manufacturing. By employing these instruments, we eliminate the estimation bias arising from potential endogenous problems, thus providing more robust and reliable results regarding the impact of digitalisation on the servicification of manufacturing in China.

In this paper, we find that digital technology adoption facilitates the servicification of manufacturing within Chinese firms by increasing the production and sale of service products. Specifically, digital technology enhances the output of all three types of services in manufacturing firms, with more prominent effects on complementary service production, such as adapting and smoothing services. However, these effects are only significant in firms that exhibit a positive attitude toward digitalisation strategy, and they are not observed in firms with either negative or neutral attitudes toward digitalisation. Further heterogeneous

analysis reveals that the positive impact of digital technology on servicification is more prominent for high-tech, innovative, upstream, state-owned, or foreign-invested manufacturing firms. These results are robust after controlling for the endogenous problem and a series of robustness checks.

Furthermore, this paper explores the possible mechanisms through which digital technology adoption facilitates the servicification process, thereby highlighting the transformative effects of digital technology on manufacturing firms' transformation into service-oriented activities. Specifically, we identify improved productivity and profitability as the key drivers behind this shift. Adopting digital technologies significantly boosts the productivity of manufacturing firms, enabling them to offer an expanded range of services. Additionally, digitalisation has a positive effect on firm profitability, providing a strong incentive for these firms to enhance their service offerings to maximise returns. These dynamics capture how digital technology adoption facilitates the integration of service activities within the manufacturing sector in China.

This paper might contribute to the literature in three aspects. First, in contrast to previous studies, this paper uses highly disaggregated product-firm-level data and provides new micro-level evidence on how manufacturing firms adopt digital technology and facilitate the output of services in production. Specifically, this paper highlights the diversity of servicification by categorising three types of product-related services in manufacturing and highlights the heterogeneity of digital technology on the servicification of firms by assessing varying attitudes toward digitalisation strategies. Second, this paper investigates the causality between digitalisation and servicification of manufacturing, providing robust evidence on the linkages between digital technology adoption and service outputs by manufacturing firms. Third, this paper proposes and examines the possible mechanisms through which digital technology affects the servicification of manufacturing, which allows for a deeper understanding of the digital transformation in manufacturing.

The rest of the paper proceeds as follows. Section 2 introduces the background and related literature. Section 3 presents the data, measurements of key variables, and empirical strategies. Section 4 describes the empirical results. Section 5 concludes and provides some policy suggestions.

2 | Background and Literature Review

Digitalisation refers to the widespread adoption of digital technologies that are used to create, process, and share information (Katz and Koutroumpis 2013). According to the boundaries of digital transformation, digitalisation could be categorised into internal and external types (Axenbeck and Breithaupt 2022). Internal digitalisation pertains to digital technology adoption within a firm or a product, whereas external digitalisation refers to digital transformation at a more macro level, such as the availability of broadband in a region.

China's digital transformation strategies have shifted focus from external digitalisation, such as improving broadband infrastructure, to internal digitalisation, which involves the digital technology adoption of firms and products. As shown in Figure 1, China's digitalisation strategies began with a strong emphasis on infrastructure development, marked by the launch of the 'Broadband China' strategy in 2013 and the subsequent establishment of digital infrastructure in cities between 2014 and 2016. The Broadband China Program paved the way for broader macroeconomic strategies at the national level, including the release of the 'Digital China Development' plan in 2018 and the formal proposal of the 'Internet Plus' strategy in 2020. Following these national-level strategies, the focus shifted to the sector level, initially prioritising the development of new digital sectors in 2021 such as automation, data analytics, the Internet of Things (IoT), and artificial intelligence (AI), etc. In 2022, the emphasis has transitioned to the digital transformation of traditional manufacturing sectors, aiming to upgrade these sectors within global value chains. The macro-level strategies have accelerated the development of digitalisation in China, which has further



FIGURE 1 | The development of China's digitalisation strategies. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/1wec.70071)]

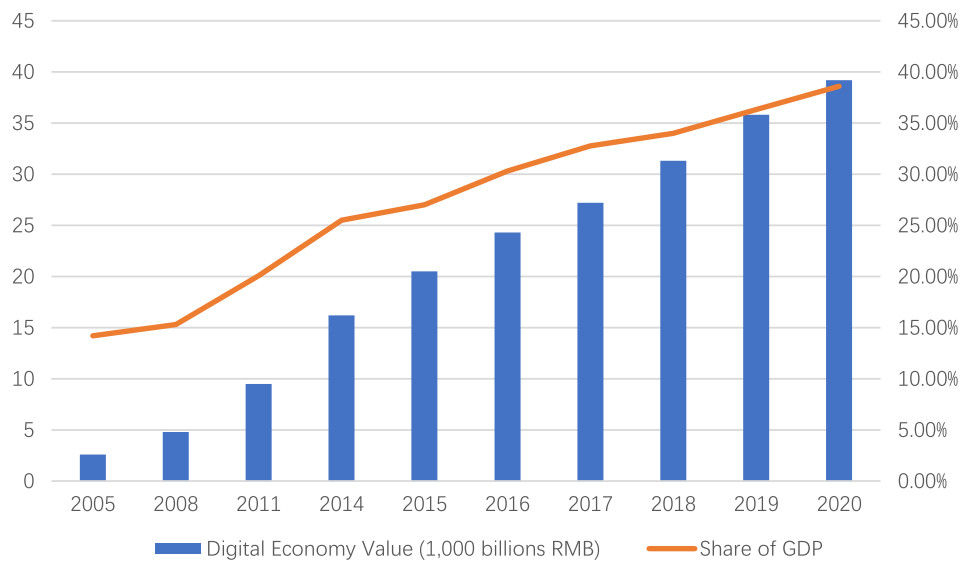


FIGURE 2 | The share of digital economy value in GDP in China. *Source:* Chinese Statistical Yearbook, compiled by the authors. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/1wec.70071)]

played a pivotal role in the Chinese economy and manufacturing upgrading. As shown in Figure 2, the value-added of digitalised sectors reached an impressive RMB39,200 billion in 2020, which accounted for a significant 38.2% of China's gross domestic product (GDP). This substantial contribution highlights the profound impact of digitalisation on the Chinese economy.

Despite the significant contribution of digitalisation to economic growth, in 2023, the Chinese government's focus shifted to the micro-level, targeting the digital transformation of individual firms, particularly small and medium-sized enterprises. The primary goal is to enhance the competitiveness and resilience of these firms in an uncertain and challenging environment characterised by sluggish economic growth (Perkins 2023), rising labour costs (Wang et al. 2020; Huang et al. 2021; Fan et al. 2021), the regionalisation of global supply chains (Los et al. 2013; Antràs and Chor 2022), and trade uncertainties (Feng et al. 2017; Liu and Ma 2020; Wang et al. 2022). To navigate these challenges, digitalisation has emerged as a preferred strategy among these firms (Li et al. 2022). More than just a technological shift (Gemini 2011), digitalisation has become an essential, comprehensive strategy in response to the growing unpredictability and complexity of the external environment, proving critical to the survival and competitiveness of these firms (Wen et al. 2022).

The rise of digitalisation marks a significant shift in firm operations. Gopalan et al. (2022) discovered that adopting ICT technologies boosts Indian firms' involvement in global value chains. Oyewo et al. (2023) found that big data analytics adoption by consulting firms can improve internal business processes, increase market share, and improve competitive position. Cardona et al. (2013) and Schweikl and Obermaier (2020) found that the digital technology represented by ICT has a positive impact on productivity, which is confirmed by Nucci et al. (2023), who found that firms investing in digital technologies from 2016 to 2018 saw their total factor productivity (TFP) grow by an average of 0.97 percentage points more than non-adopting firms. Dhyne et al. (2020) measured the digitalisation

level using digital capital stocks and found that higher digital capital correlates with an increase in a firm's value-added and profits. However, these studies mainly focus on the impacts of digitalisation on firms' performance within sectors, neglecting the impacts on firms' performance across sectors.

Chinese manufacturing firms are increasingly focusing on producing service outputs as part of their production processes, a trend known as the servicification of manufacturing (Thangavelu et al. 2018). As shown in Figure 3, the domestic service content embodied in exports has increased, whereas the foreign service content embodied in manufacturing has declined, which reflects the growing emphasis on integrating services into the manufacturing sector within China. This shift coincides with the rapid development of digital technology. However, the specific role that digitalisation plays in facilitating or accelerating this trend remains underexplored in existing literature, especially at the firm level.

The process of digitalisation has empowered Chinese manufacturing firms with services, enabling them to enhance their productivity and value-added content within GVCs (Dhyne et al. 2020; Nucci et al. 2023). Through the adoption of digital technology, firms gain access to advanced tools that facilitate the seamless integration of services. Automation, data analytics, the Internet of Things (IoT), and artificial intelligence (AI) optimise production processes, drive productivity improvements (Chen et al. 2022), and enable tailored services (Bonfanti et al. 2018). Moreover, digitalisation has fueled the emergence of innovative business models such as e-commerce and digital marketplaces, fostering solid connections between domestic service providers and manufacturers (Wang et al. 2023). For example, China's dominance in retail e-commerce transactions is evident, with a 42.4% share of the global value in 2016, surpassing the United States (US) at 33.5%. Additionally, digital platforms facilitate closer collaboration between manufacturers and service providers, strengthening global value chains and improving customer experiences (Miroudot and Cadestin 2017; Banga 2022).

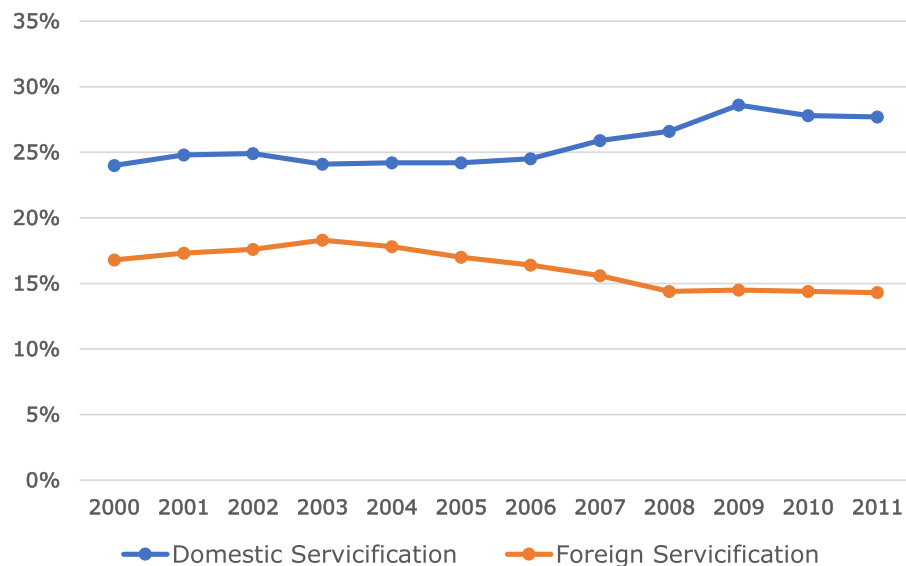


FIGURE 3 | Servicification of manufacturing in China. Source: TiVA dataset, compiled by the authors. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/tvec.12071)]

3 | Empirical Strategies and Data

3.1 | Data

The data used in this paper are derived from three main databases: the China Stock Market & Accounting Research (CSMAR) database, the Chinese Customs dataset, and Chinese cities' statistical yearbooks. These databases provide comprehensive information on various aspects related to the servicification of manufacturing, digitalisation, and relevant control variables at the firm and city levels.

The CSMAR database covers data of all listed companies on the Chinese stock market, including financial statements and information on the corporate governance of Chinese listed firms. The dataset spans from 2000 to 2020, providing a rich source of information for analysing the servicification and digitalisation measurements. By analysing the income sources of the listed firms, we can identify the extent of service-related incomes and further decompose the income sources by products. The firm-product level data enables us to capture different dimensions of servicification, such as whether the service outputs substitute or complement the manufacturing goods. We further collect the annual reports of the 2106 listed firms in China. Using the text mining approach, we capture the number of texts related to digitalisation in the annual financial reports and construct the measurement of digitalisation at the firm level.

The Chinese Customs dataset encompasses transaction-level trade data of all Chinese firms. This dataset is crucial for identifying the listed firms' performance within GVCs. By examining the trade patterns and analysing the extent of their involvement in international trade, we can assess the level of integration within GVCs for these firms, which is crucial in understanding the relationship between digitalisation, servicification, and the firms' position in GVCs.

To account for city-level control variables, we rely on Chinese cities' statistical yearbooks. These yearbooks provide a wide

range of data at the city level, including variables such as GDP per capita, education investment, and other socioeconomic indicators. Incorporating these control variables allows us to control for regional variations and capture the impact of local economic conditions on digitalisation, servicification, and GVC performance of manufacturing firms.

3.2 | The Measures of Servicification of Manufacturing Firms

Previous studies have measured the extent of servicification of manufacturing from two perspectives: the input approach and the output approach. In the input approach, servicification is measured by identifying the service inputs used by manufacturing firms and determining the share of service inputs in the total inputs (Lodefalk 2013). The use of service inputs in the manufacturing sector is often compiled from the input and output tables, which focus on the sectoral level analysis and neglect firms' heterogeneity (Pilat and Wölfl 2005; Chu and Jiang 2023). The new availability of world input-output tables also enables us to distinguish domestic service input and foreign service input, which trace the servicification of manufacturing across countries (Thangavelu et al. 2018). However, firms perform differently in the servicification process, which might lead to an estimation bias using the aggregated sectoral-level data (Lodefalk 2014).

In the output approach, servicification is measured by focusing on the service income generated by manufacturing firms (Lodefalk 2014). This approach involves identifying the income derived from service-related activities, such as maintenance, consultancy, or software support. By calculating the share of services income in the total income of manufacturing firms, we can assess the level of integration of services into their revenue streams. This measurement provides insights into how manufacturing firms have diversified their business activities to encompass service provision, thereby increasing their value-added content.

In this paper, we first measure the servicification of manufacturing from the output perspective, providing more micro-level evidence of the process of service integration into manufacturing firms. We further decompose the income source of listed firms by products, tracing the linkages between service outputs and manufacturing products. According to Cusumano et al. (2015), we have categorised the service outputs of manufacturing firms into three categories on the basis of their purposes: smoothing services, adapting services, and substituting services. Smoothing services refer to the services designed to facilitate the sale or production of manufacturing products without significantly altering their core functionalities. These services, such as financing, technical support, maintenance, and repair, aim to streamline the product lifecycle. Adapting services focus on enhancing the functionality of manufacturing products or exploring new applications, such as customisation to introduce novel features. Both smoothing and adapting services complement the core manufacturing processes. In contrast, substituting services replace traditional manufacturing products with service-oriented solutions that become the primary offerings of manufacturing firms.² This approach enables us to capture how different types of services contribute to the overall transformation of the servicification of manufacturing firms in China and investigate the impacts of digitalisation on different types of servicification.

3.3 | The Measures of Digitalisation

In this paper, we first use the text-mining approach to develop a text-based measure on Chinese listed firms' digitalisation strategies. Following Wu et al. (2021), we extract relevant information of digital technology from the annual reports to identify indicators of digitalisation. Five dimensions of digital technology are identified in this study, including AI, big data, cloud computing technology, blockchain technology, and digital technology application. Within each dimension, specific indicators are considered to assess the level of digitalisation in manufacturing firms. These indicators include 76 specific indicators, such as ICT, Virtual Private Networks (VPN), Cloud Computing, Internet of Things (IoT), 3D Printing, Robotics, etc. Figure 4 shows the five dimensions of digitalisation of Chinese manufacturing firms are increasing, suggesting a rise in digitalisation.

One challenge in employing text mining and analysis to evaluate digitalisation at the firm level lies in the intricacy of interpreting the sentiments within related texts. A positive sentiment toward digitalisation typically implies an endorsement or optimism about its potential benefits and impacts on the firm. Conversely, a negative sentiment indicates scepticism or concerns, potentially reflecting fears about the costs of digital transformation, the disruption of established processes, the risk of data breaches, or the displacement of jobs due to automation. To ensure the accuracy of digitalisation measurement, we further use sentiment analysis to evaluate the emotional tone of messages containing digitalised text, which enables us to discern the underlying attitudes toward digitalisation within firms (Shaik et al. 2023). We adopt two established dictionaries for sentiment analysis in the process: the Tsinghua University-Li Jun Chinese Commendatory Dictionary and the National Taiwan University Chinese Sentiment Polarity Dictionary. The two dictionaries

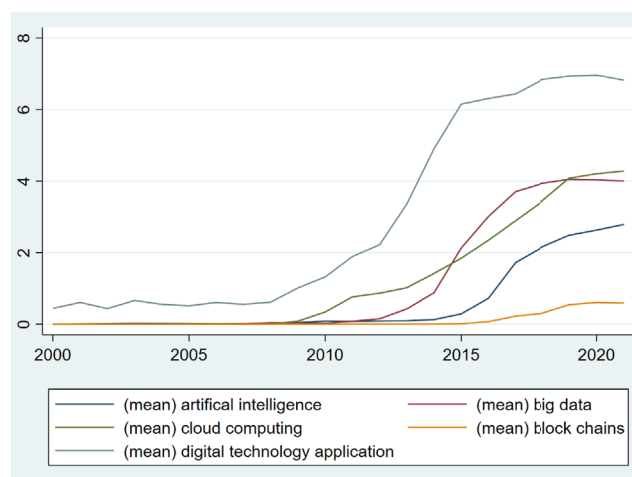


FIGURE 4 | Digitalisation of Chinese manufacturing listed firms using text mining and analysis. Source: The data are collected from the annual report of Chinese-listed manufacturing firms. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/1wec.70071)]

have discerned the sentiments associated with digitalisation texts into three categories: positive, negative, or neutral. The positive digitalisation indicators convey optimistic sentiments, which indicate a generally favorable or supportive attitude toward promoting the digitalisation of the firms. The negative sentiment reflects unfavorable emotions in the digitalisation-related text, which expresses criticism, concerns, or disagreement about the digitalisation strategy. The neutral sentence of digitalisation text contains information without emotional connotations, which usually contains objective statements or information. This approach allows us to further clarify firms' attitudes toward digitalisation, which enables us to investigate the impacts of digitalisation on the servicification of manufacturing firms.

Figure 5 illustrates the sentiment distribution (positive, negative) in digitalisation texts extracted from annual reports of Chinese manufacturing firms from 2000 to 2022. We show the average frequency of digitalisation-related text in the annual reports of listed firms across time. Following China's accession to the WTO in 2001, a notable uptrend in the frequency of digitalisation-related terms has been observed over the past three decades. The majority of Chinese manufacturing firms exhibit a positive stance toward digitalisation, evidenced by a widening gap between the frequencies of positive and negative digitalisation-related text.

Another challenge with using text-based methods to measure digitalisation lies in the inherent subjectivity of annual reports (Lee 2016). These documents are interactive and can be significantly influenced by stakeholders' preferences or opinions, potentially skewing the portrayal of a firm's digitalisation efforts (Li et al. 2022). To address this issue, we introduce another, more objective alternative indicator of digitalisation: the size of intangible assets of manufacturing firms. This measure captures the firm's actual investment in digital technologies, bridging the gap between the promises outlined in annual reports and tangible digital technology investments³ (Tambe et al. 2020).

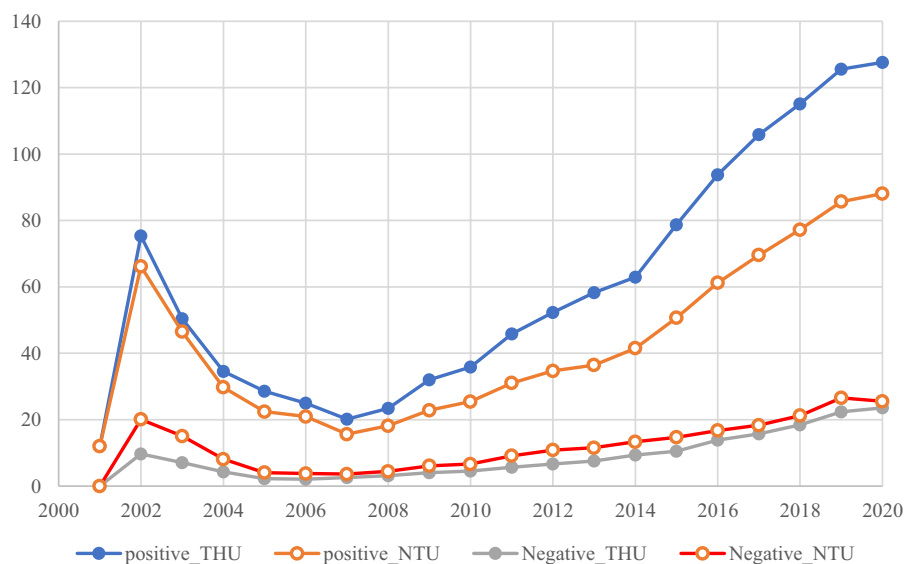


FIGURE 5 | Sentiment analysis of digitalisation texts in annual report of Chinese-listed firms. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

3.4 | Empirical Model

The model specification for examining the relationship between digitalisation and the servicification of manufacturing is represented as follows:

$$serv_{ijt} = \beta_1 digit_{ijt} + \beta_2 \mathbf{X}_{ijt} + \beta_3 \mathbf{C}_{jt} + \delta_i + \delta_t + \varepsilon_{ijt} \quad (1)$$

where i is the manufacturing listed firms, j is the city, and t is the year. $serv_{ijt}$ represents the servicification level of manufacturing for firm i in city j at time t , which captures the extent of servicification of manufacturing firms in China. $digit_{ijt}$ denotes the digitalisation level of firm i in city j at time t , which captures the digital technology adoption of this firm. $\mathbf{X}_{ijt}$ represents a vector of firm-level control variables, such as firm size, age, ownership structure, and investment in R&D. These control variables account for other factors that may affect the servicification level and allow for a more accurate estimation of the effect of digitalisation. $\mathbf{C}_{jt}$ represents a vector of city-level control variables, such as the gross domestic product (GDP) per capita, education investment, and so forth. δ_i and δ_t represent the firm and time fixed effects, respectively, which control for unobserved heterogeneity across firms and time-specific factors. ε_{ijt} is the error term. The descriptions of variables and statistical characteristics are shown in Table 1.

4 | Empirical Results

4.1 | Baseline Results

Table 2 presents the baseline estimation results for the causal relationships between servicification from income (lnservi_income) and digitalisation measures (Indigi1 and Indigi2), controlling for the firm-level and city-level control variables. Column 1 suggests that firms that adopt digital technologies and practices are more likely to generate income from service-related activities. This result is confirmed by the statistically

TABLE 1 | Variable and statistical description.

	Variable	N	Mean	SD
lnservi_income	Log of service income in total income	40,576	0.23	0.3
Indigi1	Log of digitalisation index by text-mining	40,576	1.05	1.35
Indigi2	Log of intangible asset size	40,576	2.46	1.3
lnTFPOP	Log of firm productivity	34,273	2.07	0.13
lnemp	Log of employment	40,507	7.53	1.38
age	Firm age	40,576	9.14	6.99
lnSale	Log of sales	39,852	21.17	1.57
size	Log of total asset	40,574	21.97	1.51
roa	Return to asset	40,574	0.57	117.26
roe	Return on equity	40,102	0.02	2.51

Note: TFP = total factor productivity, OP = TFP measured by the Olley and Pakes (1996) approach. The data and indicators are compiled by the authors.

significant and positive coefficient for Indigi1 (digitalisation index by text-mining), indicating that an increase in digitalisation is associated with a higher level of servicification from income. Column 2 introduces firm fixed effects, and the coefficient for Indigi1 remains significant and positive, reinforcing the finding that digitalisation positively affects servicification from income. Including the fixed effects helps control for unobserved heterogeneity across firms that may affect the outcome variable.

TABLE 2 | Baseline results.

	(1)	(2)	(3)	(4)
	Inservi_ income	Inservi_ income	Inservi_ income	Inservi_ income
Indigi1	0.031*** (0.000)	0.021*** (0.000)	0.020*** (0.000)	0.016*** (0.000)
Indigi2				0.006** (0.024)
Intfpop			0.485*** (0.000)	0.486*** (0.000)
lnemp			−0.011 (0.205)	−0.012 (0.196)
age			−0.008 (0.214)	−0.009 (0.193)
lnSale			−0.041** (0.014)	−0.042** (0.014)
size			0.029*** (0.000)	0.028*** (0.000)
roa			−0.046** (0.028)	−0.047** (0.025)
roe			−0.001*** (0.002)	−0.001*** (0.002)
Firm FE	No	Yes	Yes	Yes
Year FE	No	Yes	Yes	Yes
N	40,576	40,576	34,178	34,178
r ²	0.030	0.035	0.056	0.056

Note: p-values in parentheses; * represents $p < 0.1$, ** represents $p < 0.05$, *** represents $p < 0.01$. The results are regressed and reported by the authors.

In Column 3, we include the firm-level characteristics such as Intfpop (TFP by OP), lnemp (employment), age (firm age), lnSale (log of sales), size (log of total assets), return on assets (ROA), and return on equity (ROE) as control variables. The coefficient of digitalisation on servicification remains statistically significant and consistent with the previous models, indicating that the relationship between digitalisation and servicification from income holds even after controlling for firm-level characteristics. We also find that firms with higher total factor productivity (TFP), sales, firm size, and return on assets are associated with a higher level of servicification from income.

Column 4 further includes Indigi2 (digitalisation index by intangible asset) as an additional measure of digitalisation. The coefficient for Indigi2 is statistically significant and positive (0.006**), indicating that a higher level of digitalisation measured by intangible assets is associated with increased servicification from income. This suggests that Chinese manufacturing firms with more intangible digital assets are more likely to engage in service-oriented activities.

Overall, the baseline estimation results in Table 2 provide evidence that digitalisation, as measured by both text-mining and intangible assets, has a positive and significant impact on servicification from income in Chinese manufacturing firms. These findings suggest that adopting digital technologies and leveraging intangible assets related to digitalisation can enhance the integration of services within the manufacturing sector.

4.2 | Endogeneity

One major concern of the baseline estimation is the possible endogeneity of digitalisation. In this paper, we employed three IVs: (1) the implementation of China's Broadband policy, (2) the average digitalisation level of other firms in the same province and sector, and (3) the interaction between the number of post offices in 1984 and internet users per million people.

Firstly, we take 'China Broadband Policy' as an IV for digitalisation. The *Broadband China* policy was introduced by the Chinese government in 2013 as a top-down initiative aimed at enhancing digital infrastructure. Between 2013 and 2016, 117 cities were selected as pilot locations for its implementation. The selection process was centrally determined, making the policy largely exogenous to individual firms' decisions. The primary goal of the policy was to accelerate digital transformation by improving internet infrastructure, which is strongly correlated with firms' digitalisation levels. However, it does not directly affect firms' servicification decisions, making it a strong candidate for an instrumental variable in our analysis.

As shown in Column (1) of Table 3, we use the implementation of Broadband China as an instrument of digitalisation to evaluate the impacts of digitalisation on the servicification of manufacturing.⁴ We find that firms with higher digitalisation levels tend to produce more service output after controlling for the endogeneity, which is robust to our baseline results.

Secondly, we further use the average digitalisation level of firms in the same province and sector as a second instrumental variable to strengthen our identification strategy. This indicator is calculated on the basis of the digitalisation levels of other firms, and the individual firms' decisions have limited impacts on it, satisfying the exogeneity condition. The average digitalisation level of firms in the same province and sector is highly correlated with the digitalisation level of individual firms, as firms in the same region and industry face similar market environments and technological conditions. Meanwhile, it indirectly affects individual firms' servicification decisions by influencing their digital capabilities, satisfying the exclusion restriction. Its strength lies in capturing industry- and region-level digitalisation trends and ensuring exogenous variation that is not directly influenced by individual firms' servicification decisions. As shown in Column (2), after controlling for the endogeneity, a 1% improvement in the digitalisation level will increase the service income share in the manufacturing firm by 0.069%.

Thirdly, we use historical data as the third IV of digitalisation, which reflects past digital infrastructure development that is exogenous to current firm-level decisions, providing a strong source of variation in digitalisation while mitigating concerns

TABLE 3 | IV estimation results.

	(1)	(2)	(3)	(4)
	IV	IV	IV	IV
	Broadband China	Average digitalisation	1984 post office num	Three IVs
<i>Panel A. First-stage estimation (dependent variable: Indigitalisation1)</i>				
did	0.137*** (0.0172)			0.109*** (0.0255)
lnmean_dig1		0.116*** (0.0079)		0.113*** (0.0088)
lnint_post			−0.064*** (0.0109)	−0.0386** (0.0165)
<i>Panel B. Second-stage estimation (dependent variable: servicification index)</i>				
Indigi1	0.051** (0.040)	0.069*** (0.000)	0.083*** (0.000)	0.047*** (0.001)
Underidentification LM statistic	(63.498)***	(217.324)***	(34.460)***	(194.769)***
Overidentification <i>p</i> -value	0.000	0.000	0.000	0.2885
Weak identification F statistics	63.580	219.679	34.475	65.720
Controls	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Obs	32,142	20,802	27,480	16,305

Note: *p*-values in parentheses; * represents $p < 0.1$, ** represents $p < 0.05$, *** represents $p < 0.01$.

Source: The results are regressed and reported by the authors.

of reverse causality. In Column (3), we use the number of post offices in 1984 interacting with the internet users in the city per million people as the instrument of digitalisation. The historical infrastructure, such as the number of post offices and telephones in 1984, is a strong predictor of regional digitalisation development, as areas with better early communication networks tend to have a more advanced digital infrastructure today. Moreover, historical infrastructure was established long before firms' current digitalisation and servicification decisions, making it unlikely that these past developments were influenced by present-day firm characteristics. More importantly, since historical infrastructure primarily determined early digital connectivity rather than directly shaping firm strategies, it meets the exclusion restriction requirement. As shown in Column (3), the estimation results are still robust by using the historical indicator as the instrument, suggesting that firms with higher levels of digitalisation tend to produce more service outputs after controlling for the possible endogenous problem.

Column (4) uses all three instruments in the estimation.⁵ After controlling for endogeneity, our results remain robust to the baseline estimation, confirming that more digitalised firms tend to produce and sell more service outputs. The validity of the instruments is further supported by the first-stage estimation results, where all three IVs are statistically significant at

conventional levels, indicating strong relevance. Additionally, the weak identification test (F -statistic = 65.720) exceeds the commonly accepted threshold, suggesting that the instruments are not weak. The overidentification test yields a p -value of 0.2885, indicating that the instruments satisfy the exclusion restriction and do not introduce significant bias, reinforcing the robustness of our causal inference.

4.3 | Possible Mechanisms

This section investigates how digitalisation affects the servicification of manufacturing. In Table 4, we highlight that digital technology is a possible mechanism through which digitalisation contributes to the expansion of services within manufacturing firms. The improvement in technology is measured by the TFP by Olley and Pakes (1996). To explore this, we include the interaction between digitalisation and the TFP in the model to investigate whether digitalisation affects servicification by adjusting productivity. As shown in Column 1, the coefficient of the interaction term $\text{Indigi1} \times \text{lnTFP}$ is positive and weakly significant, suggesting that the impacts of digitalisation on the servicification of manufacturing are enhanced by productivity improvement.

In Column 2, we use an alternative approach by Levinsohn and Petrin (2003) to measure productivity. The coefficient of the

TABLE 4 | Possible mechanism: Technology effect.

	(1)	(2)	(3)	(4)
	Inservi_income	Inservi_income	Inservi_income	Inservi_income
Indigi1	−0.040 (0.243)	−0.034 (0.355)	−0.072** (0.030)	−0.082** (0.018)
Indigi2			0.006** (0.021)	0.007** (0.012)
Intfpop	0.223*** (0.000)		0.499*** (0.000)	
Indigi1 × Intfpop	0.028* (0.090)		0.042*** (0.009)	
Intfplp		0.229*** (0.000)		0.964*** (0.000)
Indigi1 × Intfplp		0.024 (0.160)		0.044*** (0.006)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
R ²	0.043	0.042	0.058	0.074
Obs	34,273	34,273	34,178	34,178

Note: *p*-values in parentheses; * represents $p < 0.1$, ** represents $p < 0.05$, *** represents $p < 0.01$. The results are regressed and reported by the authors.

interaction term Indigi1 × Intfplp becomes statistically significant and positive, suggesting manufacturing firms with higher levels of digitalisation and the TFP by Olley and Pakes (1996) are more likely to experience an increase in servicification after digitalisation.

Column 3 and Column 4 further include the alternative indicator of digitalisation (digitalisation index measured by intangible assets) in the estimation. The interaction term remains positive and statistically significant in both columns, suggesting manufacturing firms that effectively leverage digital technologies and exhibit high levels of TFP are more capable of expanding their service offerings and capitalising on the opportunities provided by new technologies. These results suggest that manufacturing firms can harness the potential of new technologies and position themselves for greater servicification, thereby enhancing their competitiveness and performance in the evolving business landscape (Acemoglu and Restrepo 2020).

Another possible channel through which digitalisation affects servicification of manufacturing sectors is profitability. On the one hand, digital transformation significantly enhances a firm's profitability. One key way is through the integration of digital technology, which further enhances firms' market competitiveness by increasing the value-added content of their products, leading to higher profitability (Cusumano et al. 2015). Moreover, digitalisation enables firms to optimise supply chain management, reduce production costs, and improve production productivity, all of which contribute to increased profitability

(Banga 2022). A compelling example is Trina Solar, a solar energy company that leveraged an industrial brain framework powered by cloud computing, IoT, big data, and deep learning. By analysing thousands of production parameters in real-time and using AI-driven optimisation, Trina Solar improved its A-grade battery cell production rate by 7%, generating millions in direct profits within just 5 months. This demonstrates how digitalisation directly enhances firms' profitability, laying the foundation for broader digital transformation and servicification.

On the other hand, firms with higher profitability have greater financial capacity to engage in service sectors. As indicated by Krugman (1979) and Melitz (2003), entering new market incurs sunk costs, by which only more profitable firms are inclined to do this. Dhyne et al. (2020) indicate that firms with strong profitability are more likely to allocate resources toward digital services to maintain market competitiveness and sustain long-term growth. Additionally, financially robust firms exhibit greater risk tolerance, allowing them to experiment with new digital service models and expand their service-oriented offerings without immediate financial constraints (Westerman et al. 2011). Furthermore, profitability enables firms to continuously enhance their servicification levels by leveraging digital tools to offer customised solutions, subscription-based services, and after-sales support, increasing both customer retention and product diversification.

As a result, profitability serves as a crucial channel through which digitalisation drives servicification. As shown in Table 4, to capture the potential mechanism of the profit effect on

servicification, we include the interactions between digitalisation (measured by *Indigi1*) with return on equity (*Lnroe*) and return on assets (*Lnroa*) to examine whether digitalisation affects the profitability of manufacturing firms, which further promotes them to produce more services.

As shown in Column 1 and Column 2, the coefficient of the interaction terms *Indigi1* × *Lnroa* and *Indigi1* × *Lnroe* are positive but not statistically significant, suggesting the joint effect of digitalisation and profitability may have a limited impact on servicification from income in this model. However, we include the alternative indicator of digitalisation in Column 3, and the coefficient of the interaction term *Indigi1* × *Lnroa* is statistically significant and positive, indicating that manufacturing firms with higher levels of digitalisation and greater returns on assets are more likely to experience an increase in servicification, potentially driven by adjustments in their profitability. In Column 4, the coefficient of *Indigi1* × *Lnroa* remains statistically significant and positive after controlling for both indicators of digitalisation, suggesting the interaction between digitalisation and return on assets plays an important role in driving servicification from income.

Overall, Table 5 underscores the significance of digitalisation and profitability indicators in shaping the servicification strategies of manufacturing firms. Firms that embrace digital transformation and achieve higher returns on assets are more likely to prioritise services as a profitable avenue, driving the evolution of their business models in response to the opportunities presented by the digital landscape.

4.4 | Robustness and Heterogeneous Analysis

4.4.1 | Sentiment Analysis of Digitalisation

As indicated above, we further distinguish the sentiment of text-based digitalisation using the sentiment analysis approach. This method enhances the robustness of research on digitalisation by incorporating the emotional and psychological dimensions of firms' digitalisation strategies. Table 6 presents the heterogeneous effects of digitalisation sentiments on the servicification of manufacturing firms. In Column (1), firms with a positive attitude toward digitalisation tend to increase their service output, as suggested by the baseline estimation. However, for firms with negative attitudes (Column 2) or neutral attitudes (Column 3) toward digitalisation, there are no significant impacts of digitalisation on the servicification of manufacturing firms. These results indicate that the ways firms view digitalisation strategies can significantly impact firms' production decisions, thereby affecting a firm's servicification process.

4.4.2 | Impacts of Different Types of Servicification

As indicated above, this paper reveals the taxonomy of services by identifying the products of manufacturing firms and categorising their service outputs into smoothing, adapting, and substituting services. In this section, we investigate the impacts of digitalisation on different types of servicification of manufacturing, providing

TABLE 5 | Possible mechanism: Profit effect.

	(1)	(2)	(3)	(4)
	Inservi_ income	Inservi_ income	Inservi_ income	Inservi_ income
<i>Indigi1</i>	0.028*** (0.000)	0.022*** (0.000)	0.026*** (0.000)	0.023*** (0.000)
<i>Indigi2</i>			0.006** (0.031)	0.006** (0.029)
<i>Lnroa</i>	−0.001 (0.756)		−0.004 (0.142)	
<i>Indigi1</i> × <i>Lnroa</i>	0.002 (0.103)		0.003** (0.021)	
<i>Lnroe</i>		0.005*** (0.007)		0.005** (0.040)
<i>Indigi1</i> × <i>Lnroe</i>		0.001 (0.478)		0.003** (0.039)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
<i>R</i> ²	0.032	0.031	0.052	0.053
Obs	36,160	36,012	30,548	30,548

Note: *p*-values in parentheses; * represents *p* < 0.1, ** represents *p* < 0.05, *** represents *p* < 0.01. The results are regressed and reported by the authors.

more detailed evidence on the role of digital technologies in transforming manufacturing processes and outputs.

Table 7 reports the heterogeneous impacts of digitalisation on different types of servicification within the manufacturing sector in China. As shown in Column (1), firms with higher levels of digitalisation tend to increase their production of smoothing services. Furthermore, digitalisation also increases the production of adapting services in manufacturing firms in Column (2). In Column (3), we also observe that digitalisation prompts firms to transition from traditional manufacturing production to service-oriented production. These results are consistent with our baseline results that digitalisation promotes the servicification of the manufacturing sector across all service categories.

4.4.3 | More Heterogenous Analysis

High-tech and low-tech manufacturing firms⁶ approach digital transformation differently, with high-tech firms generally being more adaptable to digitalisation (Sergei et al. 2023). Moreover, innovations that enhance automation and productivity, even in small-scale production, tend to adopt digital technologies more frequently and produce more complex products (Kinkel et al. 2022). Inspired by these facts, we present a more detailed

TABLE 6 | Heterogeneous effects of digitalisation by sentiment analysis.

	(1)	(2)	(3)
	Inservi_ income	Inservi_ income	Inservi_ income
positive_ Indigi1	0.042** (0.010)		
Negative_ Indigi1		0.025 (0.157)	
Neutral_ Indigi1			0.026 (0.112)
Firm FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Controls	Yes	Yes	Yes
R ²	0.029	0.026	0.026
Obs	4673	4673	4673

Note: *p*-values in parentheses; * represents $p < 0.1$, ** represents $p < 0.05$, *** represents $p < 0.01$. The variables are centered and standardised in this table.

TABLE 7 | Estimation on different types of servicification.

	(1)	(2)	(3)
	Smooth service income	Adapting service income	Substituting service income
Indigi1	0.121*** (0.000)	0.295*** (0.000)	0.041** (0.028)
Firm FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Controls	Yes	Yes	Yes
Obs	12,926	12,926	12,926

Note: *p*-values in parentheses; * represents $p < 0.1$, ** represents $p < 0.05$, *** represents $p < 0.01$.

heterogeneity analysis in this section on the basis of firms' technology levels and innovation capacities.⁷ According to the results in Columns (1) and (2) of Table 8, digitalisation has a significant positive effect on the servicification of high-tech manufacturing firms, whereas the impact on low-tech firms is not statistically significant. This pattern is further supported in Columns (3) and (4), which show that digitalisation is positively correlated with the servicification of innovative firms holding patents (i.e., more innovative firms). However, this effect does not hold for firms without patents, indicating that the relationship between digitalisation and servicification is stronger in more innovative and technologically advanced firms.

We further investigate the impact of digitalisation on the servicification of manufacturing across firms positioned at various stages

of the global value chains. The position of firms within global value chains (GVCs) might affect how digitalisation promotes their servicification processes. Upstream firms, which are involved in the earlier stages of production, primarily benefit from digital technologies that optimise production processes, improve efficiency, and streamline logistics. In contrast, downstream firms, which are closer to the end consumer, use digital tools to enhance customer experiences and improve service quality (Antràs and Chor 2022; Chor et al. 2021; Bian and Fan 2024). Following Chor et al. (2021), we measure the relative upstreamness of each firm in global value chains, which captures the firm's position within the production process and how far it is from the final consumer in terms of value-added stages. Firms with upstreamness above the median are categorised as upstream firms, whereas those with lower upstreamness are classified as downstream firms. As shown in Columns (5) and (6) of Table 8, we observe a more pronounced and statistically significant positive impact of digitalisation on the servicification of manufacturing for upstream firms within global value chains, while showing no significant effect on downstream firms. This indicates that digital technologies are more effective in enhancing service integration in upstream manufacturing firms in China, with little impact on the servicification of downstream firms.

Previous studies have highlighted the varying degrees of digitalisation across Chinese firms with different ownership structures.⁸ Although SOEs are expected to lead digital transformation efforts, they often display lower market dynamism and are more hesitant to invest the significant human, material, and financial resources necessary for such advancements (Wu et al. 2021). Private firms, on the other hand, may face greater financial constraints when adopting digital technologies (Li et al. 2023). In the last three columns of Table 8, we compare the differential impacts of digitalisation on the servicification of manufacturing across State-owned, foreign-invested, and private enterprises. The results indicate that digitalisation is positively correlated with the servicification of manufacturing in all three types of firms. However, this effect is less pronounced and statistically significant for private firms.

5 | Conclusion

This paper investigates the impacts of digitalisation on the process of servicification of manufacturing firms in China. Using the text mining and analysis approach, this paper captures the dynamic changes and quantifies the extent of digitalisation at the firm level. Using the highly disaggregated firm-product level data, this paper measures the servicification of manufacturing on the basis of income, which captures the increasing production and sales of service outputs of Chinese manufacturing firms. We find that firms with higher levels of digitalisation tend to engage more in service production, a trend that can be promoted by adopting new digital technology and improving firms' profitability. This effect is more prominent for high-tech firms, innovative firms, and upstream firms within global value chains.

Moreover, using the sentiment analysis, we further distinguish firms' attitudes toward digitalisation strategies and reveal that only firms with a positive sentiment toward digitalisation are able to improve the level of servicification in manufacturing.

TABLE 8 | More heterogenous analysis.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Low-tech	High-Tech	No patent	Patent > 1	Downstream	Upstream	SOE	FIE	Private
Indigi1	0.047 (0.305)	0.161*** (0.000)	0.041 (0.442)	0.164*** (0.000)	0.645 (0.124)	0.135*** (0.000)	0.258*** (0.000)	0.189*** (0.000)	0.067* (0.078)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Obs	5676	8033	3183	10,526	2794	10,915	6594	915	6194

Note: *p*-values in parentheses; * represents $p < 0.1$, ** represents $p < 0.05$, *** represents $p < 0.01$.

We also find that digitalisation has bolstered the production of services that complement traditional manufacturing activities as well as services that substitute manufacturing products. We also find that the positive effect of digitalisation on the servicification of manufacturing is more prominent in high-tech, innovative and upstream firms, whereas low-tech, non-innovative, and downstream firms show little to no significant impact. Additionally, private firms experience a weaker effect compared to SOEs and foreign-invested enterprises.

These findings emphasise the transformative impact of digital technology and profit-driven motivations behind the shift toward service-oriented activities within manufacturing firms. Our paper provides robust evidence that prioritising the promotion of digital transformation will not only advance the development of manufacturing sectors but also facilitate the growth of service-oriented activities, particularly in developing countries. As a result, initiatives such as providing financial incentives for technology adoption, fostering digital skills development, and creating a favourable regulatory environment for digital innovation are essential to encouraging firms to embrace digitalisation. These policy measures will not only boost manufacturing competitiveness but also help firms in developing countries achieve higher levels of servicification, ultimately transforming their role in global value chains with more service value-added content. As evidenced by studies in the tourism sector, stronger integration of ICT with financial systems can fully harness the potential of digitalisation (Adeleye et al. 2023), suggesting that similar synergies could be explored in manufacturing servicification.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

Endnotes

¹ The sentiments of texts linked to digitalization are classified into three categories: positive, negative, or neutral. Positive texts reflect optimistic views, suggesting a favourable or supportive attitude toward promoting digital transformation. Negative sentiments highlight criticism, concerns, or disagreement regarding digital strategies, while neutral sentiments provide unbiased, factual information.

² For instance, IBM's transition from hardware-centric products to a service-oriented model in the early 2000s exemplifies the substituting service shift. By realigning its business strategy to meet evolving market demands and customer preferences, IBM established itself as a prominent provider of a diversified service portfolio encompassing consulting, cloud services, cybersecurity, data analytics, and artificial intelligence.

³ A digital asset, as defined by Tambe et al. (2020), refers to a firm's investment in digital technology that yields a form of capital, which is usually intangible. This includes a broad range of digital resources and technologies that a firm invests in, such as software, databases, digital platforms, and online tools, which bolster firms' operational capabilities and digital proficiency.

⁴ In this paper, we do not directly use the Broadband China policy in our estimation but instead employ it as an instrumental variable. The main reason is that we aim to capture firm-level heterogeneity in digitalization, which a policy-level variable alone cannot fully account for. However, as suggested, we also conduct an alternative estimation by using the Broadband China Policy as a quasi-natural experiment and employ difference-in-differences (DID) approach to examine its impact on the servicification. The results are shown in Appendix A (Table A1). The results confirm that digitalization improves the servicification of manufacturing.

⁵ To ensure that our instrumental variables do not suffer from multicollinearity, we conducted a Principal Component Analysis (PCA). The results indicate that all eigenvalues were greater than 0.5 and the loading matrices showed that the variables had strong contributions in different principal components, which further supported that there was no serious multicollinearity problem among the variables. Therefore, it can be concluded that these three instrumental variables do not cause serious multicollinearity problems in the model and are suitable for simultaneous use in regression analysis.

⁶ We categorise industries into high-tech and low-tech sectors based on the “High-technology” and “knowledge-based services” aggregations based on ISIC Rev. 4¹. In this paper, we define high-tech sectors as those classified under ‘High-technology’ and ‘Medium-high-technology’

categories in the ISIC Rev. 4 classification. Meanwhile, we classify industries in the 'Medium-low-technology' and 'Low-technology' categories in the ISIC Rev. 4 classification as low-tech sectors. If the firm belongs to a high-tech sectors, the firm is defined as a high-tech firm, and low-tech firms are similarly.

⁷In this paper, we measure a firm's innovation capacity by the number of patents it has. Firms with no patents are classified as non-innovative, while those with at least one patent are categorised as innovative.

⁸The ownership of Chinese listed firms is categorised by the registration information in the CSMARs (Code of Equity Nature). We distinguish the state-owned enterprises, foreign-invested firms and private firms accordingly.

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Appendix A

Extra Robustness Check Using the Broadband China Policy

The Broadband China policy was started in 2014 and gradually implemented in 2015 and 2016 for different cities, making it suitable for constructing a time-varying difference-in-difference (DID) model. Time-varying DID modelling is a method for assessing policy or treatment effects, and is particularly suitable for situations where the timing of policy implementation varies across individuals (cities). Unlike

traditional DID models, time-varying DID models are more flexible as they allow individuals in the treatment group to receive the treatment (policy implementation) at different points in time. In this paper, we use the broadband China policy to construct a time-varying DID model to supplement the argument about the impact of digitisation on the servification of manufacturing firms. If broadbanddid = 1, it means that the city has already implemented the broadband China policy in that year; if broadbanddid = 0, it means that the city has not implemented the broadband China policy in that year. The explanatory variable is log of service income in total income, and the control variables are consistent with the baseline regression. The empirical results show that the coefficients of broadbanddids are all positive and significant in three different cases, indicating that the broadband China policy has a significant positive effect on firms' serviced income, consistent with the results using the broadband China policy as an instrumental variable.

TABLE A1 | Time-varying DID results using the broadband China policy.

	(1)	(2)	(3)
Variables	Inservi_ income	Inservi_ income	Inservi_ income
broadbanddid	0.055*** (0.003)	0.007** (0.003)	0.071*** (0.017)
lnftpop		0.539*** (0.081)	0.495*** (0.087)
lnemp		−0.006 (0.006)	−0.006 (0.006)
lnSale		−0.047*** (0.010)	−0.042*** (0.011)
size		0.031*** (0.004)	0.024*** (0.004)
roa		−0.042** (0.018)	−0.012 (0.019)
roe		−0.001*** (0.000)	−0.001*** (0.000)
Firm FE	No	Yes	Yes
Year FE	No	Yes	No
City-year FE	No	No	Yes
Sector-year FE	No	No	Yes
Controls	No	Yes	Yes
N	38,220	32,051	30,080
r ²	0.008	0.808	0.846

Note: Robust standard errors in parentheses; * represents $p < 0.1$, ** represents $p < 0.05$, *** represents $p < 0.01$. Column (1) does not add any control variables and fixed effects, Column (2) controls for control variables consistent with the baseline regression and adds two-way fixed effects for year and firm. Apart from the two-way fixed effects of Column (2), Column (3) further control the sector-year fixed effects and city-year fixed effect to eliminate the effects of city-time and sectoral-time variant variables.